

Justify your answers and show your work.

Find the limit of the following sequences or determine that the limit does not exist.

1. $\left\{ \frac{n^{10}}{3n^{10}+4} \right\}$

2. $\left\{ \frac{\cos \pi n}{\sqrt{n}} \right\}$

3. $\left\{ \frac{n!}{n^n} \right\}$

Evaluate the geometric series or explain why it diverges.

4. $\sum_{k=0}^{\infty} \left(\frac{3}{5} \right)^k$

Consider the following infinite series.

5. $\sum_{k=1}^{\infty} \frac{1}{4k^2 - 1}.$

a. Write out the first four terms of the sequence of partial sums.

b. Use the results of part (a) to propose an explicit formula for S_n .

bonus. Prove that $\sum_{k=1}^{\infty} 2k - 1 = n^2$.